

Where irrigation exists is globally contested

4. Propagation into SDG 6.4 indicators

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1 Preliminary

```
# PROPAGATION OF IRRIGATION-MAP UNCERTAINTY INTO SDG 6.4 #####
#####

# Load libraries -----

sensobol::load_packages(c("data.table", "scales", "cowplot", "ggplot2", "magrittr",
                          "here", "jsonlite", "countrycode"))

# Create custom theme -----

theme_AP <- function() {
  theme_bw() +
    theme(panel.grid.major = element_blank(),
          panel.grid.minor = element_blank(),
          legend.background = element_rect(fill = "transparent", color = NA),
          legend.key = element_rect(fill = "transparent", color = NA),
          legend.key.width = unit(0.4, "cm"),
          legend.key.height = unit(0.5, "lines"),
          legend.text = element_text(size = 6),
          legend.title = element_text(size = 7),
          axis.text.x = element_text(size = 7),
          axis.text.y = element_text(size = 7),
          axis.title.x = element_text(size = 7.3),
          axis.title.y = element_text(size = 7.3),
          axis.title = element_text(size = 10),
          plot.title = element_text(size = 8),
          strip.text.x = element_text(size = 7.4),
          strip.text.y = element_text(size = 7.4),
          strip.background = element_rect(fill = "white"),
          strip.text = element_text(margin = margin(t = 1.5, b = 1.5)))
}

# Names of the ten harmonized irrigation maps -----

maps <- c("gaez_v4", "giam", "gmia", "gripc", "luh2", "meier", "mirca2000",
          "mirca_os", "nagaraj", "spam")

# Observational subset (remote sensing / ML), used as a robustness check -----

obs_maps <- c("giam", "gripc", "gmia", "meier", "mirca2000", "mirca_os", "nagaraj")

dir.create(here("datasets", "input"), showWarnings = FALSE, recursive = TRUE)
```

2 Functions

```
# RETRIEVE OFFICIAL FAO / UN-WATER INPUTS #####

fetch_sdg_fun <- function(series_code) {
  url <- paste0("https://unstats.un.org/sdgapi/v1/sdg/Series/Data?seriesCode=",
               series_code, "&pageSize=25000")
  dt <- as.data.table(jsonlite::fromJSON(url, flatten = TRUE)$data)
  dt[, .(m49 = as.integer(geoAreaCode), area = geoAreaName,
        activity = `dimensions.Activity`, year = as.numeric(timePeriodStart),
        value = as.numeric(value))]
}

# Keep the latest year per area and activity -----

latest_fun <- function(dt) dt[!is.na(value)][order(-year), .SD[1], by = .(m49, activity)]

# World Bank mirrors AQUASTAT/national-accounts series (most-recent value) -----

fetch_wb_fun <- function(indicator) {
  url <- paste0("https://api.worldbank.org/v2/country/all/indicator/", indicator,
               "?format=json&per_page=20000&mrnev=1")
  dt <- as.data.table(jsonlite::fromJSON(url, flatten = TRUE)[[2]])
  dt[!is.na(value) & countryiso3code != "",
     .(m49 = countrycode(countryiso3code, "iso3c", "un", warn = FALSE),
       value = as.numeric(value))][!is.na(m49)]
}

# OFFICIAL 6.4.2 WATER-STRESS REPORTING BANDS (FAO / UN-Water, 5 classes) -----

stress_bands <- data.table(
  band = c("No stress", "Low", "Medium", "High", "Critical"),
  lower = c(0, 25, 50, 75, 100),
  upper = c(25, 50, 75, 100, Inf),
  order = 1:5)

classify_stress_fun <- function(x) {
  out <- rep(NA_integer_, length(x))
  for (i in seq_len(nrow(stress_bands)))
    out[x >= stress_bands$lower[i] & x < stress_bands$upper[i]] <- stress_bands$order[i]
  out
}

# PER-COUNTRY ENSEMBLE IRRIGATED AREA (m49) #####

# Aggregate the harmonized 0.2 deg field to country totals, one column per map.
```

```

ensemble_country_area_fun <- function(d) {
  dcast(d[, .(mha = sum(mha, na.rm = TRUE))], by = .(m49, dataset)],
        m49 ~ dataset, value.var = "mha")
}

# STRESS UNDER A MAP (input-swap) #####

stress_matrix_fun <- function(A, cols, ref_vec) {
  M <- sapply(cols, function(j) A$stress_nonagri + A$stress_agri * (A[[j]] / ref_vec))
  matrix(M, nrow = nrow(A), dimnames = list(NULL, cols))
}

# Inter-decile spread relative to the country median -----

interdecile_fun <- function(M) {
  md <- apply(M, 1, median, na.rm = TRUE)
  p10 <- apply(M, 1, quantile, 0.1, na.rm = TRUE)
  p90 <- apply(M, 1, quantile, 0.9, na.rm = TRUE)
  100 * (p90 - p10) / md
}

```

2.1 SDG 6.4.2 reporting bands

```

band_display <- data.table(
  Level = stress_bands$order,
  Class = stress_bands$band,
  range = c("<25$", "$25-50$", "$50-75$", "$75-100$", "$\\geq 100$"))

knitr::kable(band_display, booktabs = TRUE, escape = FALSE, linesep = "",
  label = "bands",
  col.names = c("Level", "Class", "Water stress (\\%)"),
  align = c("c", "l", "c"),
  caption = "Official SDG 6.4.2 water-stress classification")

```

Table 1: Official SDG 6.4.2 water-stress classification

Level	Class	Water stress (%)
1	No stress	< 25
2	Low	25 – 50
3	Medium	50 – 75
4	High	75 – 100
5	Critical	≥ 100

3 Retrieve and assemble the official inputs

```
# SDG 6.4.2 (level of water stress) and 6.4.1 (water-use efficiency) -----

f_642 <- here("datasets", "input", "sdg642_published.csv")
f_641 <- here("datasets", "input", "sdg641_components.csv")

if (!file.exists(f_642)) {
  s642 <- latest_fun(fetch_sdg_fun("ER_H2O_STRESS"))
  sdg642 <- dcast(s642[activity %in% c("TOTAL", "ISIC4_A01_A0210_A0322")],
                 m49 + area ~ activity, value.var = "value")
  setnames(sdg642, c("TOTAL", "ISIC4_A01_A0210_A0322"),
           c("stress_total", "stress_agri"))
  fwrite(sdg642, f_642)
}
if (!file.exists(f_641)) {
  s641 <- latest_fun(fetch_sdg_fun("ER_H2O_WUEYST"))
  sdg641 <- dcast(s641[activity %in% c("TOTAL", "ISIC4_A01_A0210_A0322")],
                 m49 + area ~ activity, value.var = "value")
  setnames(sdg641, c("TOTAL", "ISIC4_A01_A0210_A0322"), c("wue_total", "awe_agric"))
  fwrite(sdg641, f_641)
}
fwrite(stress_bands, here("datasets", "input", "sdg642_stress_bands.csv"))

sdg642 <- fread(f_642)
sdg641 <- fread(f_641)

# Agricultural value added and GDP

f_va <- here("datasets", "input", "wb_value_added.csv")
if (!file.exists(f_va)) {
  va <- fetch_wb_fun("NV.AGR.TOTL.CD"); setnames(va, "value", "gva_agr")
  gdp <- fetch_wb_fun("NY.GDP.MKTP.CD"); setnames(gdp, "value", "gdp")
  wb <- merge(va, gdp, by = "m49")
  wb[, gva_nonag := gdp - gva_agr]
  fwrite(wb, f_va)
}
wb <- fread(f_va)

# Harmonized ensemble at 0.2 deg -----

d <- fread(here("datasets", "irrigated_areas_regridded", "irrigated_areas_regridded_02.csv"))
setnames(d, "code", "m49")
cty <- ensemble_country_area_fun(d)

# Join ensemble + official inputs + agriculture share of withdrawals -----
```

```

dt_sdg <- merge(cty, sdg642[!is.na(stress_total)], by = "m49") %>%
  merge(sdg641[!is.na(wue_total), .(m49, wue_total, awe_agric)], by = "m49",
        all.x = TRUE) %>%
  merge(wb[, .(m49, gva_agr, gdp)], by = "m49", all.x = TRUE)

dt_sdg[, p_agr := fifelse(!is.na(stress_agri) & stress_total > 0,
                          stress_agri / stress_total, NA_real_)]
dt_sdg[, stress_nonagri := stress_total - fcoalesce(stress_agri, 0)]

dt_sdg <- dt_sdg[gmia > 0 & !is.na(p_agr) & stress_total > 0]
dt_sdg[, .(n_countries = .N)] %>% print()

```

```

##      n_countries
##              <int>
## 1:              170

```

4 SDG 6.4.2: water-stress propagation

```

stress_cols <- paste0("stress_", maps)
band_cols   <- paste0("band_",   maps)

M_gmia <- stress_matrix_fun(dt_sdg, maps, dt_sdg$gmia)
for (j in maps) dt_sdg[[paste0("stress_", j)]] <- M_gmia[, j]
for (j in maps) dt_sdg[[paste0("band_",   j)]] <- classify_stress_fun(M_gmia[, j])

dt_sdg[, band_base := classify_stress_fun(stress_total)]
dt_sdg[, band_min  := apply(as.matrix(.SD), 1, min, na.rm = TRUE), .SDcols = band_cols]
dt_sdg[, band_max  := apply(as.matrix(.SD), 1, max, na.rm = TRUE), .SDcols = band_cols]
dt_sdg[, span      := band_max - band_min]
dt_sdg[, stress_idr := interdecile_fun(M_gmia)]

# how many countries change reporting band across the ten maps -----

cat("Countries crossing >= 1 stress band:",
    dt_sdg[span >= 1, .N], sprintf("(%.0f%%)", 100 * dt_sdg[, mean(span >= 1)]), "\n")

## Countries crossing >= 1 stress band: 71 (42%)

cat("Countries crossing >= 2 stress bands:",
    dt_sdg[span >= 2, .N], sprintf("(%.0f%%)", 100 * dt_sdg[, mean(span >= 2)]), "\n")

## Countries crossing >= 2 stress bands: 30 (18%)

# Observational-only ensemble (drops scenario/allocation products) -

M_obs <- stress_matrix_fun(dt_sdg, obs_maps, dt_sdg$gmia)
B_obs <- apply(M_obs, 2, classify_stress_fun)
obs_span <- apply(B_obs, 1, max, na.rm = TRUE) - apply(B_obs, 1, min, na.rm = TRUE)
cat("Observational-only (7 maps), crossing >= 1 band:",
    sum(obs_span >= 1), sprintf("(%.0f%%)\n", 100 * mean(obs_span >= 1)))

## Observational-only (7 maps), crossing >= 1 band: 66 (39%)

# Which map supplies the extreme stress-----

cat("\nMap most often supplying the MAXIMUM stress:\n")

##
## Map most often supplying the MAXIMUM stress:
print(sort(table(maps[apply(M_gmia, 1, which.max)]), decreasing = TRUE))

##
##   nagaraj   meier   giam   gripc   gaez_v4   luh2   spam mirca_os
##      44      33      28      24      13      13      11      4

```

```
# CROSSINGS OF THE NO-STRESS / STRESS LINE (25%) #####
```

```
band_label <- stress_bands$band
```

```
# Officially safe but pushed into stress by >= 1 valid map -----
```

```
up <- dt_sdg[band_base == 1 & band_max >= 2]
up[, n_stress := rowSums(.SD >= 25), .SDcols = stress_cols]
cat("No stress -> stress (officially 'No stress', flipped by >= 1 map):",
    nrow(up), "\n")
```

```
## No stress -> stress (officially 'No stress', flipped by >= 1 map): 24
```

```
print(up[order(-band_max)][, .(area, worst = band_label[band_max],
    n_stress, n_total = length(maps))][1:10])
```

```
##           area      worst n_stress n_total
##           <char>    <char>    <num>   <int>
##  1:      Somalia Critical         6      10
##  2:         Cuba      High         8      10
##  3:         Chad   Medium         1      10
##  4:   Argentina      Low         1      10
##  5:     Burundi      Low         1      10
##  6:     Czechia      Low         1      10
##  7: El Salvador      Low         1      10
##  8:      Greece      Low         1      10
##  9:     Grenada      Low         1      10
## 10:      Kenya      Low         1      10
```

```
# Officially stressed but declared safe by >= 1 valid map -----
```

```
down <- dt_sdg[band_base >= 2 & band_min == 1]
down[, n_nostress := rowSums(.SD < 25), .SDcols = stress_cols]
cat("\nStress -> No stress (officially stressed, declared safe by >= 1 map):",
    nrow(down), "\n")
```

```
##
```

```
## Stress -> No stress (officially stressed, declared safe by >= 1 map): 16
```

```
print(down[order(-band_base, -n_nostress)][, .(area, published = band_label[band_base],
    n_nostress, n_total = length(maps))][1:10])
```

```
##           area published n_nostress n_total
##           <char>    <char>    <num>   <int>
##  1:      Sri Lanka      High         1      10
##  2:   Afghanistan   Medium         1      10
##  3:      Indonesia      Low         6      10
##  4:         Cyprus      Low         4      10
##  5:   Dominican Republic      Low         4      10
##  6:      Philippines      Low         4      10
```



```

## 7: Timor-Leste Low 4 10
## 8: Ethiopia Low 3 10
## 9: Japan Low 3 10
## 10: Democratic People's Republic of Korea Low 3 10
# k-OF-10 CONSENSUS SWEEP #####

cell <- d[, .(n_present = sum(mha > 0), mbar = mean(mha), m49 = m49[1]),
  by = .(lon, lat)]

k_sweep <- rbindlist(lapply(1:length(maps), function(k) {
  cell[n_present >= k, .(area_k = sum(mbar)), by = m49][, k := k]
})) %>%
  merge(cty[, .(m49, gmia)], by = "m49") %>%
  merge(dt_sdg[, .(m49, area, stress_nonagri, stress_agri, band_base)], by = "m49")

k_sweep[, stress_k := stress_nonagri + stress_agri * (area_k / gmia)]
k_sweep[, band_k := classify_stress_fun(stress_k)]

k_cross <- k_sweep[, .(span = max(band_k, na.rm = TRUE) - min(band_k, na.rm = TRUE)),
  by = .(m49, area)][span >= 1]
cat("Countries changing band across the k-sweep:", nrow(k_cross), "\n")

## Countries changing band across the k-sweep: 40

cat("At near-consensus k = 5, differing from the published band:",
  k_sweep[k == 5 & band_k != band_base, .N], "\n")

## At near-consensus k = 5, differing from the published band: 13

```

5 SDG 6.4.1: water-use-efficiency propagation

```
# FULL SECTORAL NUMERIC #####

dt_wue <- dt_sdg[!is.na(wue_total) & !is.na(awe_agric)]
ref_area <- dt_wue$gmia # GMIA reference (same as 6.4.2): r = area_map / area_GMIA
swe_eff <- (dt_wue$wue_total - dt_wue$awe_agric * dt_wue$p_agr) / (1 - dt_wue$p_agr)

W <- sapply(maps, function(j) {
  r <- dt_wue[[j]] / ref_area
  pa_map <- (dt_wue$p_agr * r) / (dt_wue$p_agr * r + (1 - dt_wue$p_agr))
  dt_wue$awe_agric * pa_map + swe_eff * (1 - pa_map)
})
dt_wue[, wue_idr := interdecile_fun(W)]

cat("SDG 6.4.1 countries analysed:", nrow(dt_wue), "\n")

## SDG 6.4.1 countries analysed: 162

cat("Inter-decile WUE spread: median",
    sprintf("%.0f%%", dt_wue[, median(wue_idr, na.rm = TRUE)]),
    "of baseline; within +/-25%:",
    sprintf("%.0f%% of countries\n", 100 * dt_wue[, mean(wue_idr <= 25, na.rm = TRUE)]))

## Inter-decile WUE spread: median 67% of baseline; within +/-25%: 16% of countries

cat("Magnitude vs 6.4.2 (inter-decile median): 6.4.1",
    sprintf("%.0f%%", dt_wue[, median(wue_idr, na.rm = TRUE)]), "vs 6.4.2",
    sprintf("%.0f%%", dt_sdg[, median(stress_idr, na.rm = TRUE)]),
    "-> comparable in magnitude.\n")

## Magnitude vs 6.4.2 (inter-decile median): 6.4.1 67% vs 6.4.2 74% -> comparable in magnitude

cat("Asymmetry is categorical: only 6.4.2 has reporting bands, so its movement",
    "reclassifies countries; 6.4.1 is a continuous USD/m3 with no band system.\n")

## Asymmetry is categorical: only 6.4.2 has reporting bands, so its movement reclassifies coun

# Persist the propagated results -----

fwrite(dt_sdg[, c("m49", "area", "stress_total", "band_base", "p_agr",
                 stress_cols, "band_min", "band_max", "span", "stress_idr"),
        with = FALSE],
        here("datasets", "output", "sdg642_ensemble.csv"))
fwrite(dt_wue[, .(m49, area, wue_total, awe_agric, p_agr, wue_idr)],
        here("datasets", "output", "sdg641_full_numeric.csv"))
```

5.1 Figure: the two indicators are equally map-dependent

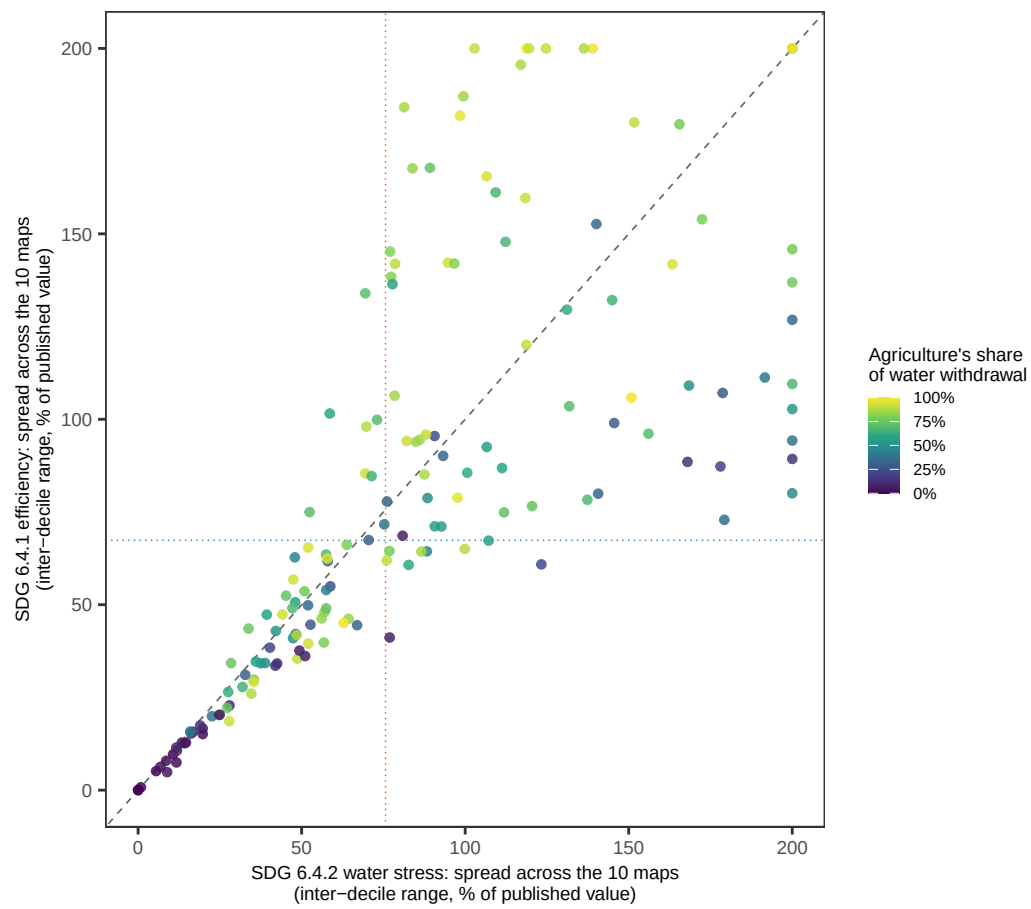
```
# SDG 6.4.1 vs 6.4.2 SENSITIVITY #####

ax_cap <- 200

plot_efficiency <- ggplot(dt_wue, aes(stress_idr, wue_idr, colour = p_agr)) +
  geom_abline(slope = 1, intercept = 0, linetype = 2,
             linewidth = 0.3, colour = "grey40") +
  geom_hline(yintercept = median(dt_wue$wue_idr, na.rm = TRUE),
            linetype = 3, linewidth = 0.25, colour = "#2c7fb8") +
  geom_vline(xintercept = median(dt_wue$stress_idr, na.rm = TRUE),
            linetype = 3, linewidth = 0.25, colour = "#d95f0e") +
  geom_point(size = 1.2, alpha = 0.85) +
  scale_colour_viridis_c("Agriculture's share\nof water withdrawal",
                       limits = c(0, 1), labels = label_percent()) +
  scale_x_continuous("SDG 6.4.2 water stress: spread across the 10 maps\n(inter-decile range, %",
                    limits = c(0, ax_cap), oob = squish) +
  scale_y_continuous("SDG 6.4.1 efficiency: spread across the 10 maps\n(inter-decile range, %",
                    limits = c(0, ax_cap), oob = squish) +
  theme_AP() + theme(legend.position = "right")

ggsave(here("datasets", "output", "fig_sdg641_efficiency.pdf"),
       plot_efficiency, width = 5.5, height = 4.8)

plot_efficiency
```



6 Composite figure

```
pal_band <- c("No stress" = "#1a9850", "Low" = "#a6d96a", "Medium" = "#fee08b",
             "High" = "#fc8d59", "Critical" = "#d73027")

x_cap <- 150
rng <- copy(dt_sdg)
rng[, st_min := apply(as.matrix(.SD), 1, min, na.rm = TRUE), .SDcols = stress_cols]
rng[, st_max := apply(as.matrix(.SD), 1, max, na.rm = TRUE), .SDcols = stress_cols]
rng[, rank   := frank(stress_total, ties.method = "first")]

band_rect <- data.table(xmin = c(0, 25, 50, 75, 100),
                       xmax = c(25, 50, 75, 100, x_cap),
                       band_f = factor(band_label, levels = band_label))

plot_range <- ggplot() +
  geom_rect(data = band_rect,
            aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = nrow(rng) + 1,
                fill = band_f), alpha = 0.30) +
  geom_linerange(data = rng, aes(y = rank, xmin = pmin(st_min, x_cap),
                                xmax = pmin(st_max, x_cap)),
                 linewidth = 0.25, colour = "grey25") +
  geom_point(data = rng, aes(y = rank, x = pmin(stress_total, x_cap)),
             size = 0.45, colour = "black") +
  scale_fill_manual("Stress band", values = pal_band, drop = FALSE) +
  scale_x_continuous("Water stress (%)", limits = c(0, x_cap),
                    oob = squish, expand = c(0, 0)) +
  scale_y_continuous("Countries (ranked by published water stress)",
                    expand = c(0, 0)) +
  theme_AP() +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(),
        legend.position = "none")

map_label <- c(gmia = "GMIA (FAO)", gaez_v4 = "GAEZ+", giam = "GIAM",
              gripc = "GRIPC", luh2 = "LUH2", meier = "Meier",
              mirca2000 = "MIRCA2000", mirca_os = "MIRCA-OS",
              nagaraj = "Nagaraj", spam = "SPAM")

swap <- melt(dt_sdg[span >= 2], id.vars = c("area", "stress_total"),
            measure.vars = band_cols, variable.name = "map", value.name = "band")
swap[, map := sub("band_", "", map)]
swap[, map_f := factor(map_label[map],
                      levels = c("GMIA (FAO)", sort(map_label[names(map_label) != "gmia"])))
swap[, area := factor(area, levels = dt_sdg[span >= 2][order(stress_total), area])]
swap[, band_f := factor(band_label[band], levels = band_label)]

plot_swap <- ggplot(swap, aes(map_f, area, fill = band_f)) +
```

```

geom_tile(color = "white", linewidth = 0.3) +
scale_fill_manual("Stress band", values = pal_band, drop = FALSE) +
scale_x_discrete(position = "top") +
labs(x = NULL, y = NULL) +
theme_AP() +
theme(axis.text.x = element_text(angle = 45, hjust = 0, size = 7),
      axis.text.y = element_text(size = 7),
      legend.position = "right")

# Combine -----

plot_sdg <- plot_grid(plot_range, plot_swap, ncol = 2,
                      rel_widths = c(0.35, 0.65), labels = "auto")

# Print last so the caption attaches to the figure float -----

plot_sdg

```

